

HelixCode — Open Issues Tracker

Per Constitution §11.4.15 (Item-status tracking) + §11.4.16 (Item-type tracking) + §11.4.19 (Fixed-document column-alignment) + CONST-057 (Type-aware closure vocabulary) + CONST-058 (Reopened-source attribution).

Authoritative resumption ledger: docs/CONTINUATION.md (CONST-044). This file complements it with item-level granularity for currently-open work.

Status vocabulary (closed set): Queued | In progress | Ready for testing | In testing | Reopened | Fixed/Implemented/Completed (→ Fixed.md)

Type vocabulary (closed set): Bug | Feature | Task

Prefix convention

Round 189 (2026-05-19) introduced per-project / per-submodule ID prefixes replacing the legacy ISSUE-NNN flat namespace. New items **MUST** use the prefix matching their scope; cross-cutting items affecting two or more submodules (or any meta-repo concern) live under HXC. Numeric portion is per-prefix (each prefix starts at 001). Legacy ISSUE-NNN IDs renamed forward-only per CONST-043; historical close-out narratives in docs/CONTINUATION.md preserve original IDs verbatim, and docs/Fixed.md annotates each closure with (ex-ISSUE-NNN) for git-history traceability.

Prefix	Scope	Source
HXC	HelixCode root project (project-wide, multi-submodule, governance, infrastructure)	this repo
HXA	HelixAgent submodule	dependencies/ HelixDevelopment/ HelixAgent (when

Prefix	Scope	Source
		present) / helix_agent/ tree
HXM	HelixMemory submodule	dependencies/ HelixDevelopment/ HelixMemory
HXL	HelixLLM submodule	dependencies/ HelixDevelopment/ HelixLLM
HXQ	HelixQA submodule	dependencies/ HelixDevelopment/ HelixQA (helix_qa/)
HXS	HelixSpecifier submodule	dependencies/ HelixDevelopment/ HelixSpecifier
HXO	HelixOrchestrator (= LLMOrchestrator) submodule	dependencies/ HelixDevelopment/ LLMOrchestrator
HXV	HelixVerifier (= LLMsVerifier) submodule	dependencies/ HelixDevelopment/ LLMsVerifier
HXD	HelixDocProcessor (= DocProcessor) submodule	dependencies/ HelixDevelopment/ DocProcessor
HXI	HelixI18n (when added)	tba
PLN	Planning submodule (vasic- digital)	dependencies/vasic- digital/Planning
VEN	VisionEngine submodule (HelixDevelopment)	dependencies/ HelixDevelopment/ VisionEngine
SLF	SelfImprove submodule (vasic- digital)	dependencies/vasic- digital/SelfImprove
STO	Storage submodule (vasic-digital)	dependencies/vasic- digital/Storage
OPS	LLMOps submodule (vasic- digital)	dependencies/vasic- digital/LLMOps
VDB	VectorDB submodule (vasic- digital)	dependencies/vasic- digital/VectorDB
OBS		

Prefix	Scope	Source
	Observability submodule (vasic-digital)	dependencies/vasic-digital/Observability
MCP	MCP_Module submodule (vasic-digital)	dependencies/vasic-digital/MCP_Module
MSG	Messaging submodule (vasic-digital)	dependencies/vasic-digital/Messaging
MDW	Middleware submodule (vasic-digital)	dependencies/vasic-digital/Middleware
PLG	Plugins submodule (vasic-digital)	dependencies/vasic-digital/Plugins
STR	Streaming submodule (vasic-digital)	dependencies/vasic-digital/Streaming
WAT	Watcher submodule (vasic-digital)	dependencies/vasic-digital/Watcher
CNV	conversation submodule (vasic-digital)	dependencies/vasic-digital/conversation
AUT	Auth submodule (vasic-digital)	dependencies/vasic-digital/Auth
LZY	Lazy submodule (vasic-digital)	dependencies/vasic-digital/Lazy
ATP	AutoTemp submodule (vasic-digital)	dependencies/vasic-digital/auto_temp
PLI	PliniusCommon submodule (vasic-digital)	dependencies/vasic-digital/plinius_common
CHL	challenges submodule (vasic-digital)	challenges/
CNT	containers submodule	containers/
SEC	security submodule	security/
PAN	panoptic submodule	panoptic/

For submodules not listed above, default to the first 3 letters of the submodule name, uppercase (e.g. Watcher → WAT). Document the new prefix in this table on first use.

Legacy → new mapping (round 189)

Old ID	New ID	Scope rationale
ISSUE-001	VEN-001	VisionEngine helix-gitlab URL
ISSUE-002	VEN-002	VisionEngine vasic-digital-github fork divergent
ISSUE-003	HXL-001	HelixLLM analysis_test.go hardcoded path
ISSUE-004	HXL-002	HelixLLM TOON WriteT00N 500
ISSUE-005	HXC-001	CONST-052 rename programme (project-wide)
ISSUE-006	HXC-002	Round-74 residual LOGIC FAILs (multi-submodule cross-cutting)
ISSUE-007	HXC-003	CONST-046 migration backlog (project-wide)
ISSUE-008	HXQ-001	helix_qa TestPerformance flake
ISSUE-009	HXA-001	helix_agent handler tests (4)
ISSUE-010	HXA-002	helix_agent debate API drift
ISSUE-011	HXA-003	venice CONST-037 (in helix_agent)

VEN-001 (ex-ISSUE-001) — VisionEngine helix-gitlab remote repo missing (404) — CLOSED (→ Fixed.md)

Status: Completed (→ Fixed.md) **Type:** Task **Discovered:** 2026-05-19 (round 98 — Planning + VisionEngine i18n migration)
Discovered-By: AI subagent during 4-remote push attempt
Closed-By: Round 188 (subagent repo-inventory sweep) **Root cause:** The helix-gitlab remote URL in dependencies/HelixDevelopment/VisionEngine/.git/config pointed at git@gitlab.com:HelixDevelopment/visionengine.git — a non-existent group path. The actual GitLab group is helixdevelopment1 (path) / HelixDevelopment (display name). The repository helixdevelopment1/VisionEngine (id 80411994) already existed since 2026-03-19. NOT a missing-repo issue — a URL-misconfiguration issue. **Fix:** git remote set-url helix-gitlab git@gitlab.com:helixdevelopment1/VisionEngine.git in the VisionEngine submodule, then git push helix-gitlab master (FF-safe: local was 46 commits ahead, remote 0 ahead). Push landed at SHA 2d0c35b (verified via git ls-remote helix-gitlab master). The Upstreams recipe push-helix-gitlab.sh references the remote by name (not URL), so it continues to work unchanged. **Evidence:** - glab api projects/helixdevelopment1%2Fvisionengine → id 80411994 OK -

git ls-remote helix-gitlab HEAD →
2d0c35bebb199a9a199fbf899eaeb292e38eaf17 (matches local HEAD) -
Original broken URL still 404s when probed directly (proves URL
was the issue, not perms)

HXL-001 (ex-ISSUE-003) — HelixLLM internal/agents/tools/analysis_test.go hardcoded absolute path

Status: Fixed (→ Fixed.md) **Type:** Bug **Discovered:** 2026-05-19
(round 95 — HelixLLM migration; surfaced as pre-existing failure)
Discovered-By: AI subagent during HelixLLM standalone test
run **Closed-By:** Round 105 (commit a5e56d4 in HelixLLM; meta
pointer fedd152) **Attribution correction:** Originally documented
as helix_agent; actual location is HelixLLM submodule
(dependencies/HelixDevelopment/HelixLLM/internal/agents/tools/).
Commit SHAs 0a84310 resolved there. **Resolution:** Replaced
hardcoded path with t.TempDir() + 2 synthesised fixture files.
Bonus: same bug-pattern discovered in git_test.go (constant
helixLLMRoot + 7 tests) — refactored gitSandbox() signature. 6 tests
now PASS on any host. Mutation verified.

HXL-002 (ex-ISSUE-004) — HelixLLM internal/gateway/middleware TOON WriteT00N returns 500

Status: Fixed (→ Fixed.md) **Type:** Bug **Discovered:** 2026-05-19
(round 95) **Discovered-By:** AI subagent **Closed-By:** Round 105
(commit a5e56d4) **Attribution correction:** Originally documented
as helix_agent; actual location is HelixLLM submodule. Commit
6f11c56 resolved there. **Resolution:** Root cause was vasic-digital/
TOON's round-27 anti-bluff change (Marshal returns
ErrT00NEncodingNotImplemented unconditionally) combined with
WriteT00N treating ANY Marshal error as 500. Fix: fall back to
json.Marshal while preserving application/toon Content-Type
(matches ContentNegotiation middleware). 500 still returned for
genuinely unmarshallable values (channels). 19 middleware tests
now PASS. Mutation verified.

HXC-001 (ex-ISSUE-005) — CONST-052

rename programme: meta-repo

directories still PascalCase

Status: In progress **Type:** Task **Discovered:** 2026-05-15 (CONST-052 cascade landed) **Discovered-By:** Constitution **Evidence:** Meta-repo top-level dirs already snake_case (round-88 sweep). Remaining non-compliance is two layers deeper: dependencies/HelixDevelopment/* + dependencies/vasic-digital/* owned-org submodule dirs (PascalCase), and 59 Upstreams/ dirs inside submodule trees. **Resolution path:** Phased migration per CONST-052 §11.4.29. Round 113 produced the phased plan (f666410, docs/superpowers/specs/2026-05-19-const052-rename-programme-plan.md). Round 343 executed the safe (zero-submodule-go.mod-entanglement) batches.

Round-343 12 chosen snake_case names (operator “agent defaults”): D-1 sequential phases; D-2 helix_development (parent dir, deferred — touches every consumer go.mod); D-3 vasic-digital kept (GitHub-org handle, proper-noun carve-out); D-4 n/a (helix_code already snake_case); D-5 LLMsVerifier/Assets+Website deferred (deployment-wire audit); D-6 mcp_module; D-7 i_llm; D-8 toon; D-9 rag; D-10 cluster-C upstreams strict; D-11 yes co-authored; D-12 one approval per batch.

Round-343 per-batch status:

Batch	Renamed	Status	Evidence
1	HelixDevelopment/ Models → models	LANDED alea3c8	submodule resolves; go build ./ internal/... ./ cmd/... exit 0
2	HelixDevelopment/ DebateOrchestrator → debate_orchestrator 11 vasic-digital/* zero-go.mod- consumer leaves (auto_temp, claritas, doc_processor, gandalf_solutions, hyper_tune, i_llm, leak_hub, ouroborous, plinius_common, veritas, vision_engine)	LANDED 416fe8e	go list -m digital.vasic.debate → new path; build exit 0
3		LANDED e813b5c	11 submodule statuses resolve; build exit 0

Batch	Renamed	Status	Evidence
Deferred	~37 owned-org leaves consumed by helix_agent/ helix_qa/HelixLLM go.mod	DEFERRED	renaming requires submodule-internal go.mod commits entangled with pre- existing uncommitted work — needs dedicated per-submodule rounds
Deferred	parent dirs HelixDevelopment/ →helix_development/ (D-2), vasic- digital/ kept (D-3)	DEFERRED	parent rename touches every consumer go.mod atomically
Deferred	59 Upstreams/ →upstreams/ (cluster C, D-10)	DEFERRED	live inside submodule trees — separate-repo commits

13 of ~58 owned-org leaf renames done this round (zero build breakage). HXC-001 stays In progress pending the deferred submodule-entangled and parent-dir batches.

HXC-002 (ex-ISSUE-006) — Round-74 residual LOGIC-class FAILs (CLOSED)

Status: Fixed (→ Fixed.md) **Type:** Bug **Discovered:** 2026-05-19 (round 74 — release-gate-test.sh creation; classified by round 89) **Discovered-By:** AI release-gate sweep **Closure progress:** - ✓ **HelixMemory:** closed round 106 (commit 69016df — single-line go.mod fix; 6 FAIL → 0 FAIL) - ✓ **vasic-digital/Planning:** round 107 NO-OP — 275 PASS / 0 FAIL / 20 SKIP-OK; likely incidentally fixed by round 98 i18n migration - ✓ **helix_agent inner:** closed round 109 (commit 0f492e98 — 5 test-side bluff fixes, zero production changes) **Evidence:** Round 74 surfaced 26 FAILs across submodules; rounds 82-87 closed 19; this Issue tracked the residual 7 across 3 submodules. All 3 components closed by rounds 106 + 107 + 109. **Follow-ups surfaced (NEW issues filed):** 4 helix_agent handler tests previously masked by mid-run panic (now visible) + 3 build-failed packages depending on sibling submodule API drift (digital.vasic.debate) + 2 LOGIC FAILs reclassified as cross-cutting work (venice CONST-037 model-wiring + compliance CONST-051 architectural reconciliation). See HXA-001 through HXA-003 (filed below).

HXA-001 (ex-ISSUE-009) — helix_agent handler tests surfaced after round-109 fix

Status: Queued **Type:** Bug **Discovered:** 2026-05-19 (round 109)
Discovered-By: AI subagent (helix_agent LOGIC audit)
Evidence: Mid-run panic in TestIsProviderAvailable_NotAvailable aborted test binary; round 109's fix unblocked execution, surfacing 4 pre-existing FAILs:
TestFormattersHandler_FormatCode_UnsupportedLanguage,
TestEmbeddingHandler_WithRealManager, TestGetTaskResources,
TestGetTaskLogs. Out of round-109's 5-fix cap. **Resolution path:**
Per-handler investigation, similar to round 109's test-side bluff pattern.

HXA-002 (ex-ISSUE-010) — helix_agent debate/llmprovider sibling-submodule API drift

Status: Fixed (→ Fixed.md) **Type:** Bug **Discovered:** 2026-05-19 (round 109) **Discovered-By:** AI subagent **Investigated:** 2026-05-20 (round 324) — split into a mechanical part and a design-decision part. **Closed:** 2026-05-20 (round 342) **Closure-Ref:** helix_agent commit (round-342 HXA-002 debate API drift) + meta-repo .gitmodules pointer-bump **Investigation finding (operator's explicit ask — moved vs deleted):** The learning/knowledge/recommendations capability tier was **genuinely DELETED, not moved**. git log on the digital.vasic.debate submodule (dependencies/HelixDevelopment/debate_orchestrator, renamed from DebateOrchestrator per CONST-052) shows the orchestrator was rebuilt from scratch — commit 196d0ea "feat: initial DebateOrchestrator reconstruction (Phase 1)". orchestrator/api.go has carried only the slim CreateDebate/GetStatistics surface since that very first commit (git log --follow orchestrator/api.go = single entry). A tree-wide grep of dependencies/ for KnowledgeRepository, GetRecommendations, ConvertAPIRequest, GetDebateStatus, DefaultMinConsensus, MaxAgentsPerDebate, EnableAgentDiversity found **zero** surviving copies in any digital.vasic.* package or in HelixSpecifier/HelixMemory. The slim API is the first and only version — the richer tier was a pre-reconstruction artifact that no longer exists anywhere. Per the operator's chosen direction for the deleted case, the helix_agent tests were rewritten down to the slim API. **Resolution:** See docs/Fixed.md row for the full closure narrative (Part-1 import swap + Part-2 slim-API rewrite + score-scale + go.mod rename-drift fix + captured evidence).

HXA-003 (ex-ISSUE-011) — venice TestGetCapabilities model-list drift (CONST-037)

Status: Fixed (→ Fixed.md) **Type:** Bug **Discovered:** 2026-05-19 (round 109) **Discovered-By:** AI subagent **Closed:** 2026-05-19 (round 190) **Closure-Ref:** helix_agent commit (round-190 venice CONST-037 model-list drift) + meta-repo pointer-bump **Evidence:** Test hardcoded venice-uncensored; Venice API returned 75 models with the family rotated to venice-uncensored-1-2 / venice-uncensored-role-play. Per CONST-037 (LLMsVerifier is the single source of truth for model metadata) the assertion violated the no-hardcoded-list rule. **Resolution:** helix_agent/internal/llm/providers/venice/venice_test.go::TestGetCapabilities — replaced `assert.Contains(..., "venice-uncensored")` and `assert.Contains(..., "llama-3.3-70b")` with structural assertion: `NotEmpty(SupportedModels)` plus a substring scan for the `venice-uncensored*` family. SKIP-OK marker per CONST-035 fires if the entire family disappears (avoids false-positive PASS). Mutation-verified (revert → FAIL with the original drift, restore → PASS).

HXC-004 — Recovery-batch under- verification (40% FAIL rate per round-193 audit)

Status: Fixed (→ Fixed.md) **Type:** Bug **Discovered:** 2026-05-19 (round 193 — recovery-batch verification audit) **Discovered-By:** AI subagent **Fixed:** 2026-05-19 (round 200 — per-package test-assertion repair) **Evidence:** Round-193 audit of 10 recovery-batch-landed packages (recovery commits b7f8672 + 5c94696) found 6 PASS / 4 FAIL: - internal/llm (round 161): test-assertion drift — tests still expected pre-i18n English literal “api_key”, production emits message-ID `internal_llm_wizard_anthropic_apikey_required` - internal/logo (round 163): same drift — tests expected “failed to open” / “failed to decode”, production emits `internal_logo_open_source_failed` / `internal_logo_decode_source_failed` - internal/notification (round 167): same drift — tests expected Title literals “Task Completed”, “Task Failed”, “Workflow Completed”, “Workflow Failed”, “Worker Disconnected”, “System Error”, “System Started”; production emits `internal_notification_title_*` IDs - internal/performance (round 168): build break — `translator.go` `stdctx.Context` vs plain context — fixed inline by parent agent **Root cause:** Recovery-batch commits captured stalled-agent file content but did NOT re-run consuming-test updates + did NOT verify build/test green per-package. **Resolution:** Round-200 subagent updated test assertions in all 3 drifted packages to expect message-ID echoes

(internal_<pkg>_* prefix). Per-package PASS confirmed (llm: 51.8s, logo: 0.07s, notification: 0.89s, performance: 8.4s). Per CONST-035 mutation-verified one assertion per package: revert to literal → FAIL (production emits the ID), restore → PASS. **Audit reference:** docs/audits/2026-05-19-recovery-batch-verification.md (commit 1badef1).

HXC-003 (ex-ISSUE-007) — CONST-046 migration backlog (57,329 violations baselined; shrinking)

Status: In progress **Type:** Feature **Discovered:** 2026-05-19 (round 92 — audit script) **Discovered-By:** AI subagent ground-truth scan **Evidence:** Round-92 scan reported 57,345 violations across 21,937 files. Round 99b baseline collapsed to 54,803 unique (path, literal_hash) keys. Phase 4 (rounds 100+) systematically migrating top-concentration files: round 100 (evaluators.go), 101 (challenge_recorded_ai testgen.go), 102 (challenge_desktop.go) — see CONTINUATION.md close-outs. **Resolution path:** Continued Phase 4 cadence; audit-gate --fail-on-new already enforced; each migration round MUST re-run --update-baseline so snapshot shrinks toward zero.

HXQ-001 (ex-ISSUE-008) — helix_qa intermittent TestPerformance flake (host-load-sensitive)

Status: Fixed (→ Fixed.md) **Type:** Bug **Discovered:** 2026-05-19 (round 82) **Discovered-By:** AI subagent **Fixed:** 2026-05-20 (round 325) **Evidence:** helix_qa TestPerformance (three perf tests in pkg/vision/ — TestPerformance_DHash64_Under5msPer1080pFrame, TestPerformance_PHash_Under25msPer1080pFrame, TestPerformance_SSIM_Under5msPer480pFrame) fails intermittently under high host load (concurrent containers + builds). Not a code bug per se; a sensitivity issue — the hard per-frame timing ceilings (5 ms / 25 ms / 5 ms) are only meaningful on a quiescent host. **Resolution:** Decision — **path (b)** (env-var gating) chosen over path (a) (loosen tolerance). Rationale: loosening the timing tolerance would weaken the test's anti-bluff value — a genuine perf regression could then pass. Path (b) preserves the strict assertions while making the flake deterministic: the three tests now check os.Getenv("HOST_LOAD_DEDICATED") and t.Skip("SKIP-OK: #HXQ-001 ...") honestly when unset, running strict only on a quiescent dedicated host (HOST_LOAD_DEDICATED=1). This is the CONST-035-compliant choice. Landed in helix_qa submodule

commit 649e2dd + meta .gitmodules pointer bump. docs/test-coverage.md §6.1 documents the env var. Post-fix evidence: go build ./pkg/vision/... exit 0, go vet clean; go test -count=1 -run TestPerformance ./pkg/vision/... (unset) → all 3 --- SKIP with SKIP-OK: #HXQ-001 marker; HOST_LOAD_DEDICATED=1 go test -count=1 -run TestPerformance ./pkg/vision/... → all 3 --- PASS strict (DHash64 average 741ns, PHash average 88.969µs — well under the 5 ms / 25 ms ceilings).

HXC-005 — cmd/ performance_optimization_standalone/main.go is a CONST-035 simulation bluff

Status: Fixed (→ Fixed.md) **Type:** Bug **Discovered:** 2026-05-20 (round 317 — cmd i18n migration subagent) **Discovered-By:** AI subagent — refused to localize the file because doing so would polish a bluff **Fixed:** 2026-05-20 (round 318) **Evidence:** helix_code/cmd/performance_optimization_standalone/main.go was a package main that printed “ Starting HelixCode Production Performance Optimization” then *simulated* every optimization phase: // Simulate production optimization phases, time.Sleep(500 * time.Millisecond) per phase, and improvement := 5.0 + rand.Float64()*20.0 — fabricated improvement percentages from a random number generator. No real profiling, no real optimization, no real measurement. The canonical BLUFF-001-class anti-pattern (CLAUDE.md §3.3 / §6 ANTI-PATTERN 1) — a binary that reports success for work it never performed. Violated CONST-035 / Article XI §11.9. **Resolution:** Decision — **DELETE** (resolution path b). The standalone tool was genuinely obsolete: fully superseded by cmd/performance_optimization/ (snake_case post-CONST-052), which calls the REAL dev.helix.code/internal/performance.PerformanceOptimizer — real runtime.ReadMemStats, real GOMAXPROCS tuning, real before/after measurement — and carries CONST-046 i18n + a unit-test file. git rm -r cmd/performance_optimization_standalone/ removed the dead bluff; stale references purged from docs/COMPREHENSIVE_AUDIT_REPORT.md. Reproduce-before-fix Challenge added at cmd/performance_optimization/bluff_regression_test.go: TestHXC005_BluffStandaloneDirectoryDeleted asserts the obsolete path is gone + the real command survives; TestHXC005_RealOptimizerMeasuresActualMemory allocates a retained 32 MiB buffer and asserts the optimizer’s baseline MemoryUsage tracks a genuine runtime.HeapAlloc reading (not an RNG single-digit). Post-fix evidence: go build ./cmd/... exit 0; go test -count=1 -run TestHXC005 ./cmd/performance_optimization/ → both PASS (literal log: optimizer baseline MemoryUsage=33812624 bytes, runtime.HeapAlloc=33802008 bytes — both real measurements,

same order of magnitude. No RNG-fabricated improvement percentages.); anti-bluff smoke on the deleted path returns N/A (directory gone).

PAN-001 — panoptic appendString truncates multi-byte UTF-8 runes to bytes (TestResult.MarshalJSON corrupts non-ASCII)

Status: Fixed (→ Fixed.md) **Type:** Bug **Discovered:** 2026-05-19 (round 298 — panoptic enrichment subagent) **Discovered-By:** AI subagent runner-detector against real executor.TestResult.MarshalJSON **Fixed:** 2026-05-19 (round 302 — panoptic submodule commit 24aa627 + meta pointer bump) **Evidence:** panoptic/internal/executor/executor.go:120 — buf = append(buf, byte(r)) in the else branch of appendString casts a rune to a single byte. Multi-byte UTF-8 codepoints (German umlauts, Spanish accents, Japanese CJK, Serbian Cyrillic, Chinese Han) get truncated to one byte each, producing corrupted JSON output. Honestly tracked via the round-298 Challenge runner's executor-marshal:utf8-detector:regression-present PASS line + KNOWN-ISSUE entry in panoptic/docs/test-coverage.md §7. Affects every consumer that JSON-marshals a TestResult containing non-ASCII text. **Resolution:** Replaced buf = append(buf, byte(r)) with buf = utf8.AppendRune(buf, r) (Go 1.21+) + added unicode/utf8 import. Single-line functional fix. Post-fix evidence: go test -race -count=1 ./internal/executor/... → ok 4.470s; bash challenges/panoptic_describe_challenge.sh → 39/39 PASS, 0 FAIL; runner UTF-8 detector flipped from regression-present → fixed (literal log: PASS [executor-marshal:utf8-detector:fixed] + KNOWN-ISSUE-RESOLVED: executor.appendJSONString now UTF-8 clean). Closed in this round.

HXV-002 — LLMsVerifier verification/package 10 pre-existing test failures

Status: Fixed (→ Fixed.md) **Type:** Bug **Discovered:** 2026-05-20 (round 345 — LLMsVerifier i18n round-12 subagent) **Discovered-By:** AI subagent — git stash test confirmed the 10 failures reproduce at submodule HEAD 582ae9c7 (round-336) *without* the round-345 i18n change, proving pre-existing and unrelated **Resolved:** 2026-05-20 (round 348) **Evidence:** go test ./verification/... in dependencies/HelixDevelopment/LLMsVerifier/llm-verifier/ reported 10 failures; after round-348 fix ok digital.vasic.llmsverifier/verification (1.635s), 0 failures, go

build ./... clean. **Resolution:** All 10 failures classified **(A) test-assertion drift** — every failing test asserted pre-honesty fabricated behaviour that round-17 commit a6328629 correctly removed. **No production code changed.** Per-failure classification table:

#	Test	File
1	TestVerifier_Verify_Success	verification.go
2	TestVerifier_Verify_ResultScores	verification.go
3	TestVerifier_Verify_LatencyMetrics	verification.go
4	TestVerifier_Verify_CodeLanguageSupport	verification.go
5	TestVerifier_Verify_CodeCapabilities	verification.go
6	TestVerifier_Verify_ModelStatusFlags	verification.go
7	TestVerifier_Verify_ContextCancellation	verification.go
8	TestVerifier_Verify_MultipleRequests	verification.go
9	TestCodeVerificationService_TestCodeVisibility_Error	code_verification.go
10	TestCodeVerificationService_VerifyModelCodeVisibility_ServerError	code_verification.go

Mirrors HXV-001 round-323’s classification approach. The production code (verification.go round-17 ErrVerificationNotWired, code_verification.go error propagation + zero-response→failed) was already honest; only the stale test assertions needed re-keying to certify it.

HXC-006 — HelixCode Speed Programme — CLOSED (migrated to docs/Fixed.md)

Status: Implemented (→ Fixed.md) — see docs/Fixed.md for the full closure record. **Type:** Feature All 6 phases / 31 tasks landed; CONST-048 coverage ledger at docs/research/speed/05-coverage-ledger.md (29 PASS + 2 PARTIAL + 0 DEFERRED). Closed by P5-T04 round 400. Section retained as a migration tombstone per §11.4.19 — the authoritative closure narrative is in docs/Fixed.md.

HXC-011 — helix_qa runner emits hollow sub-microsecond “PASSED” metadata rows for desktop-platform bank cases

Status: Queued **Type:** Bug **Discovered:** 2026-05-20 (speed-programme HelixQA bank registration) **Discovered-By:** AI subagent (speed-programme P5-T04 close-out — surfaced during Phase 0-5 HelixQA bank registration) **Evidence:** The helix_qa autonomous runner emits PASSED rows for test-bank cases on the desktop platform with sub-microsecond reported durations — durations physically impossible for a case that actually executed (no process spawn, no I/O, no real workload could complete that fast). The runner records the metadata PASSED row without ever executing the case. This is a §11.4 PASS-bluff in the QA runner itself (Article XI §11.9 — a PASS that carries no positive runtime evidence). Pre-existing; not introduced by the speed programme — surfaced while registering the speed-programme banks for the desktop platform. **Resolution path:** Audit the helix_qa runner’s per-platform dispatch for the desktop target; the case-execution path must actually invoke the bank case and capture real wire evidence (process exit code, stdout/stderr, duration ≥ a plausible floor) before emitting PASSED. A case that cannot be executed on the desktop platform MUST report SKIP-OK with a marker — never a hollow PASSED. Closure requires a reproduce-before-fix test that plants a known-failing case and asserts the desktop dispatch reports FAIL/SKIP-OK (never PASSED) when the case did not really run. Cascaded into the helix_qa submodule per CONST-047.

HXC-012 — data race in helix_code/internal/llm/load_balancer.go background stat-collector goroutine

Status: Queued **Type:** Bug **Discovered:** 2026-05-20 (speed-programme -race test floors) **Discovered-By:** AI subagent (speed-programme Phase 2/3 -race runs) **Evidence:** A data race in helix_code/internal/llm/load_balancer.go — the background stat-collector goroutine reads/writes shared load-balancer statistics without synchronisation, and surfaces under full-package parallel go test -race of internal/llm. Pre-existing; not introduced by the speed programme — surfaced while running the Phase 2/3 -race test floors mandated by the per-task test template (04-phased-implementation-plan.md §10). Reported honestly: the speed-programme tasks that touched internal/llm (P1-T01..07, P3-T01) ran -race clean on the *specific units they changed*; the race is in the pre-existing stat-collector path, which no speed task modified. **Resolution path:** Guard the shared stat-collector state with a sync.Mutex/sync.RWMutex (or convert the counters to sync/atomic), per the CLAUDE.md §4.2 manager pattern. Closure requires a reproduce-before-fix test that runs the load balancer under -race with concurrent stat collection and FAILS pre-fix, PASSES post-fix. Composes with P4-T04 (profile-gated contention tuning) — the fix should not regress the contention profile.

HXC-007 — Constitution §11.4.68/70-74 cascade + meta-pointer bump

Status: In progress **Type:** Task **Discovered:** 2026-05-20 (constitution-submodule pull) **Discovered-By:** AI subagent (CONST-049 fetch-before-edit pipeline) **Evidence:** Per CONST-049 the constitution submodule was fetched + pulled first (584b3ee → 34a82b3). The pull carried 6 new rules (§11.4.68 + §11.4.70-74). All 6 cascaded — verbatim or by ID reference per CONST-047 — into the meta-repo governance files and 67 owned-submodule CONSTITUTION.md / CLAUDE.md / AGENTS.md. Meta-repo .gitmodules constitution pointer bumped to the new HEAD. **Resolution path:** Cascade itself is complete; this item stays In progress because the CONST-049 step-4 multi-upstream reconciliation (push to every configured upstream of every cascaded submodule) is running concurrently and the GitHub ↔ GitLab mirrors are still converging. Closes to Completed (→ Fixed.md) once every cascaded submodule's mirrors are confirmed at parity. Tracked alongside HXC-009 (mirror-divergence reconciliation).

HXC-008 — CONST-055 G1 governance gaps surfaced by post-constitution-pull validation sweep

Status: In progress **Type:** Bug **Discovered:** 2026-05-20 (CONST-055 post-pull validation sweep, gate G1) **Discovered-By:** AI subagent (scripts/verify-all-constitution-rules.sh run after the §11.4.68/70-74 pull) **Evidence:** The CONST-055 mandatory post-constitution-pull sweep surfaced two pre-existing governance gaps (NOT regressions introduced by the round's cascade work): - (a) scripts/verify-all-constitution-rules.sh references a non-existent dependencies/HelixDevelopment/Models path — stale verifier config. The actual on-disk directories are lowercase dependencies/HelixDevelopment/models (post-CONST-052 rename, batch 1, commit a1ea3c8) plus dependencies/vasic-digital/Models. The verifier config drifted out of sync with the rename. - (b) helix_qa/CONSTITUTION.md is missing the older anchors CONST-047 through CONST-057; dependencies/HelixDevelopment/VisionEngine/CONSTITUTION.md is missing §11.4.69. **Resolution path:** (a) update the verifier's path reference to the lowercase models directory; (b) cascade the missing CONST-047..057 anchors into helix_qa/CONSTITUTION.md and §11.4.69 into VisionEngine's CONSTITUTION.md per CONST-047. Both gaps are pre-existing cascade-drift defects; closure requires a re-run of scripts/verify-all-constitution-rules.sh reporting G1 clean.

HXC-009 — Owned-submodule GitHub ↔ GitLab mirror-divergence reconciliation

Status: In progress **Type:** Task **Discovered:** 2026-05-20 (multi-upstream push during §11.4.68/70-74 cascade) **Discovered-By:** AI subagent (4-remote push attempt during HXC-007 cascade) **Evidence:** Systemic lineage divergence between the vasic-digital ↔ HelixDevelopment GitHub/GitLab mirrors of several owned submodules — the org mirrors of the same logical repository had drifted apart (each carrying commits absent from the other), so a plain push to all upstreams would be non-FF. helix_qa already reconciled via a merge-first 2-parent commit a0397a4. VisionEngine, LLMPProvider, and other owned submodules are still under reconciliation. **Resolution path:** Per CONST-061 / §11.4.71 merge-first discipline — real 2-parent merge commits preserving the union of both lineages, NO force-push, NO history rewrite, NO branch deletion. Each reconciled submodule pushes the merge to every configured upstream once FF-safe. Closes once every

divergent owned submodule's mirrors are confirmed at parity. Composes with HXC-007 (the cascade push that surfaced the divergence).

HXC-010 — End-to-end Kimi CLI + Qwen Code CodeGraph verification blocked by LLM backend quota/credentials

Status: Operator-blocked **Type:** Task **Discovered:** 2026-05-20 (CodeGraph incorporation Phase C close-out — caveat 1 re-attempt) **Discovered-By:** AI subagent (tools/codegraph/challenges/cg-challenge-05-kimi.sh + cg-challenge-07-qwen.sh re-run) **Evidence:** Both per-agent Challenges were re-driven non-interactively this session against the real CodeGraph MCP server (624,092-node graph of the HelixCode repo). Neither agent could be driven to a true end-to-end answer because their LLM backends are blocked: - **Kimi CLI** — MCP transport works fully: the transcript (docs/research/codegraph/evidence/phase-c/CG-CHALLENGE-05-kimi-transcript.txt) shows Kimi's MCP loader connecting to the codegraph server and enumerating all 9 codegraph_* tools (codegraph_search, codegraph_context, codegraph_callers, codegraph_callees, codegraph_impact, codegraph_node, codegraph_explore, codegraph_status, codegraph_files). The agent step then aborts with Error code: 429 – You've reached kimi monthly usage limit for this billing cycle. Connect-only + tool-discovery PASS captured; end-to-end NOT proven. - **Qwen Code** — the transcript (docs/research/codegraph/evidence/phase-c/CG-CHALLENGE-07-qwen-transcript.txt) shows Qwen OAuth free tier was discontinued on 2026-04-15. Run /auth to switch to Coding Plan, OpenRouter, Fireworks AI, or another provider. Connect-only PASS captured (config registers codegraph MCP server + binary reachable); end-to-end NOT proven. This is reported HONESTLY per §11.4.3 — the connect-only fallback is never claimed as end-to-end. Claude Code, OpenCode, and Crush remain true-end-to-end verified (Phase C). **Operator-Block-Details:** WHAT: true end-to-end CodeGraph verification for Kimi CLI and Qwen Code (each agent invokes a codegraph_* tool and returns real graph data in its transcript). WHY: every agent-side self-resolution path is exhausted — Kimi's backend returns HTTP 429 monthly-quota-exceeded, Qwen's bundled OAuth free tier was discontinued upstream on 2026-04-15; neither can be unblocked without paid/credentialed access the agent does not possess. UNBLOCK CONDITION: operator supplies (a) Kimi API quota/credentials with remaining billing-cycle budget AND/OR (b) Qwen Code LLM credentials (Coding Plan, OpenRouter, Fireworks AI, or another supported provider via qwen /auth); then re-run tools/codegraph/challenges/

cg-challenge-05-kimi.sh + cg-challenge-07-qwen.sh. WHO: operator.
Resolution path: Closes to Completed (→ Fixed.md) once the operator supplies the credentials and the two Challenges are re-run producing tier-1 PASS (a real .go symbol path in each transcript). Until then the connect-only + tool-discovery evidence is the strongest honest proof and is captured under docs/research/codegraph/evidence/phase-c/.

Last regenerated: 2026-05-20 (round 400 — speed-programme close-out: HXC-006 closed Implemented (→ Fixed.md) after all 31 tasks landed + CONST-048 coverage ledger docs/research/speed/05-coverage-ledger.md; HXC-011 + HXC-012 filed — two pre-existing defects surfaced during the programme: helix_qa desktop-platform hollow-PASS runner bluff + internal/llm/load_balancer.go stat-collector data race). Previous round 399 — HXC-010 filed: Kimi/Qwen end-to-end CodeGraph verification operator-blocked. To update Issues_Summary.md mechanically, run scripts/generate_issues_summary.sh (TODO: create — currently this Issues.md is the source of truth and Summary is hand-maintained).